DIGITAL MEDIA CONTENT PUBLISHING MATRIX

VidHub – A DSA Based Video Streaming Platform

PROJECT REPORT

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1 Student Details

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Subject	Data Structures and Algorithms
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Language Used	C++

2 Project Title

Digital Media Content Publishing Matrix

Project Name: VidHub – A DSA Based Video Streaming Platform

3 Problem Statement

Modern video streaming platforms process millions of videos every day. Managing large video catalogs, tracking watch history, verifying copyright licenses, ranking content, generating recommendations, and optimizing delivery routes require efficient data structures and algorithms.

The objective of VidHub is to create a simplified streaming platform that demonstrates how data structures such as Binary Search Trees, Stacks, Queues, Hash Maps, Heaps, Graphs, and Deques can be used to solve real-world streaming problems efficiently.

4 Objectives

- Maintain an organized catalog of videos.
- Track recently watched videos.
- Generate sequential recommendations.
- Verify copyright licenses quickly.
- Rank videos using engagement scores.
- Model content sharing networks.
- Find optimal delivery routes.
- Simulate smooth video streaming.

5 DesignandArchitecture

Mo dule	Data Structure / Algorithm
Video Catalog	Binary Search Tree
Watch History	Stack
Recommendation Queue	Queue
License Registry	Hash Map
Ranking System	Max Heap
Sharing Network	Graph
Delivery Optimizer	Dijkstra Algorithm
Video Segment Manager	Deque

The VidHub platform is divided into multiple modules, each responsible for solving a specific problem. Appropriate data structures were selected to ensure efficient storage, searching, ranking, recommendation generation, and content delivery.

6 AlgorithmDescription

6.1 VideoCatalog(BST)

Stores videos in sorted order and supports efficient insertion and searching.

6.2 WatchHistory(Stack)

Maintains recently watched videos using Last In First Out (LIFO) behavior.

6.3 Recommendation Queue

Processes recommendations using First In First Out (FIFO) ordering.

6.4 License Registry

Uses Hash Map lookups for instant license verification.

6.5 Ranking System

Uses Max Heap to display top-ranked videos.

6.6 Sharing Network

Represents content distribution using Graphs.

6.7 DeliveryOptimizer

Uses Dijkstra's Algorithm to determine minimum-cost delivery routes.

6.8 VideoSegmentManager

Uses Deque to manage video chunks efficiently.

7 Complexity Analysis

Op eration	Complexity
Video Search (BST)	O(log n)
Video Insert (BST)	O(log n)
Stack Push	O(1)
Queue Insert	O(1)
License Verification	O(1)
Heap Insert	O(log n)
Graph Traversal	O(V+E) O(E
Dijkstra Algorithm	log V) O(1)
Deque Operations	

8 Screenshots of Execution

8.1 Login Screen

```
○ ayush@Ayushs-MacBook-Air-5 VidHub % ./VidHub

   ╔══════════════════════════════════════════════════════════════╗
   ║         🎬  VidHub  v1.0.0  —  DSA Video Platform            ║
   ╚══════════════════════════════════════════════════════════════╝


   ==========================================================
    🔑  Select a Session (Login As)
   ==========================================================

   --- Viewers ---
   [1]   [USR-001] Riya Sharma          | Mumbai       | viewer
   [2]   [USR-002] Kabir Singh          | Delhi        | viewer
   [3]   [USR-006] Meera Patel          | Pune         | viewer
   [4]   [USR-007] Rohan Verma          | Hyderabad    | viewer
   [5]   [USR-008] Ananya Rao           | Chennai      | viewer

   --- Creators ---
   [6]   [USR-003] CodeWithArjun        | Uploads: 7   | Subscribers: 12400
   [7]   [USR-004] PriyaWellness        | Uploads: 3   | Subscribers: 8900
   [8]   [USR-009] TechWorldHub         | Uploads: 0   | Subscribers: 340

   --- Admin ---
   [9]   [USR-005] Admin Kumar          | System       | *** ADMIN ***

   Select session [1-9]: █
```

Figure 1: Application Login Interface

8.2 Creator Dashboard

Figure 2: Creator Dashboard

8.3 License Verification

```
✅  Logged in as Creator: CodeWithArjun

==========================================================
  🎬  Creator Menu
==========================================================
[1] My Dashboard
[2] Upload a Video
[3] Delete a Video
[4] Home Feed (Top Videos)
[5] Browse Catalog (A-Z)
[6] Watch a Video
[7] Watch History
[8] Share a Video
[9] Switch Session
[0] Exit VidHub

> 2

==========================================================
  📤  Upload New Video
==========================================================
Title      : Pythonn
Genre      : TECH
License Key: CC-EDU-001
Duration (sec) [30-7200]: 4500

❌  License "CC-EDU-001" is invalid or revoked. Upload rejected.
```

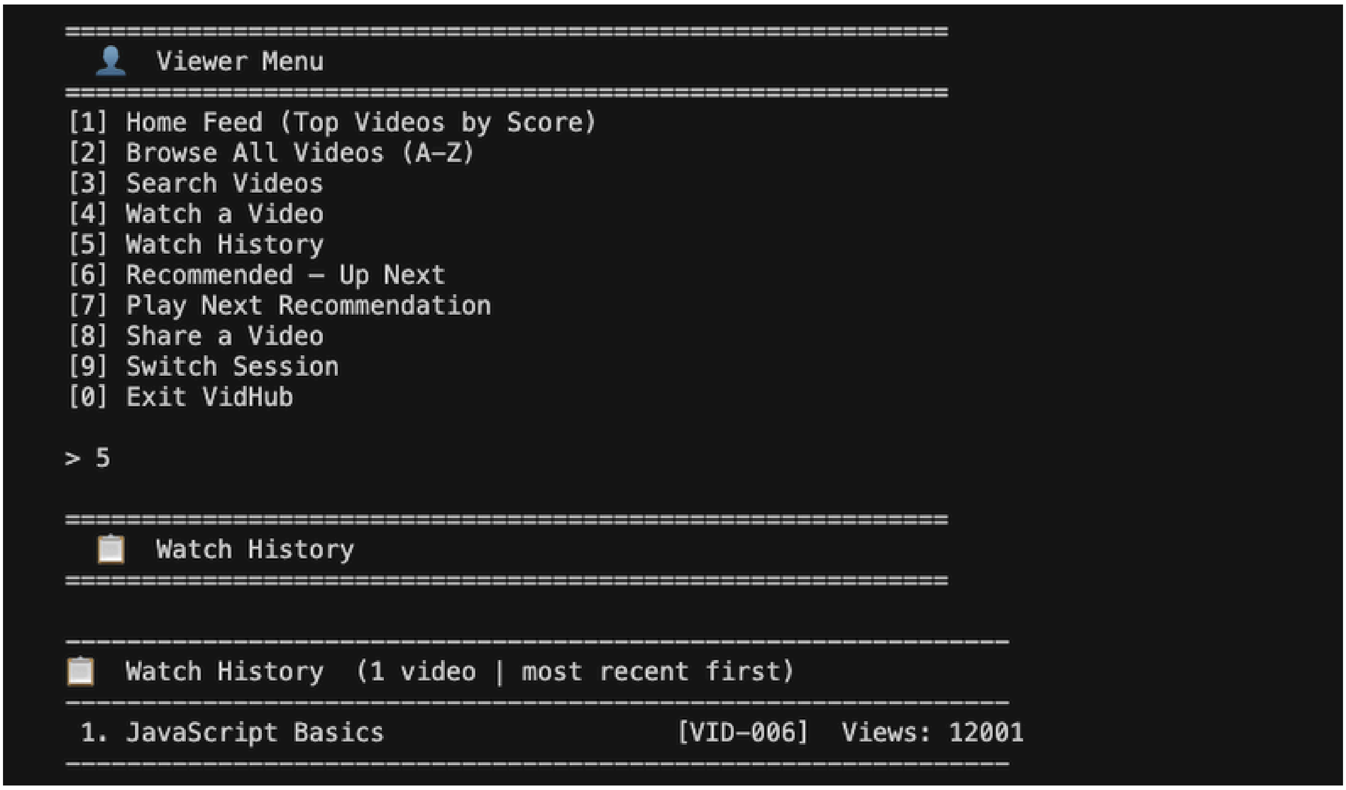
Figure 3: License Validation

8.4 VideoCatalog

```
✅  Logged in as Viewer: Kabir Singh

==========================================================
  👤  Viewer Menu
==========================================================
[1] Home Feed (Top Videos by Score)
[2] Browse All Videos (A-Z)
[3] Search Videos
[4] Watch a Video
[5] Watch History
[6] Recommended — Up Next
[7] Play Next Recommendation
[8] Share a Video
[9] Switch Session
[0] Exit VidHub

> 2

==========================================================
  📼  All Videos (A-Z)
==========================================================
screenshot4.png

==================================================================
📼  Video Catalog  (10 videos | alphabetical order)
==================================================================
  1. Build a REST API                      [VID-003]  Tech        Score: 53911.1
  2. Guitar Chords for Beginners           [VID-009]  Music       Score: 17188.9
  3. JavaScript Basics                     [VID-006]  Education   Score: 11960.0
  4. Morning Yoga Routine                  [VID-004]  Health      Score: 56400.0
  5. Python OOP Crash Course               [VID-002]  Education   Score: 18593.7
  6. Python for Beginners                  [VID-001]  Education   Score: 24133.3
  7. React.js Full Course                  [VID-010]  Tech        Score: 44235.2
  8. Stock Market for Dummies              [VID-008]  Finance     Score: 36633.3
  9. Top 10 Travel Hacks                   [VID-005]  Travel      Score: 54550.0
 10. Vegan Cooking 101                     [VID-007]  Lifestyle   Score: 16011.7
==================================================================
```

Figure 4: Video Catalog

8.5 WatchHistory



```
==========================================================
  👤  Viewer Menu
==========================================================
[1] Home Feed (Top Videos by Score)
[2] Browse All Videos (A-Z)
[3] Search Videos
[4] Watch a Video
[5] Watch History
[6] Recommended — Up Next
[7] Play Next Recommendation
[8] Share a Video
[9] Switch Session
[0] Exit VidHub

> 5

==========================================================
  📋  Watch History
==========================================================

--------------------------------------------------------------
📋  Watch History  (1 video | most recent first)
--------------------------------------------------------------
 1. JavaScript Basics                   [VID-006]  Views: 12001
--------------------------------------------------------------
```

Figure 5: Watch History

9 Results and Findings

The project successfully demonstrates the practical implementation of Data Structures and Algorithms in a video streaming environment.

Key achievements include:

- Efficient content organization using BST.
- Quick watch history tracking using Stack.
- Ordered recommendation delivery using Queue.
- Fast copyright verification using Hash Maps.
- Ranking through Max Heap.

- Network modelling through Graphs.
- Shortest path computation using Dijkstra Algorithm.
- Efficient chunk management using Deque.

10 Conclusion

VidHub successfully simulates the core functionalities of a modern video streaming platform. By integrating Binary Search Trees, Stacks, Queues, Hash Maps, Heaps, Graphs, Dijkstra's Algorithm, and Deques, the system achieves efficient content management, recommendation generation, copyright verification, network optimization, and streaming supp ort.

11 GitHubRepositoryLink

https://github.com/Ayusshh020/VidHub